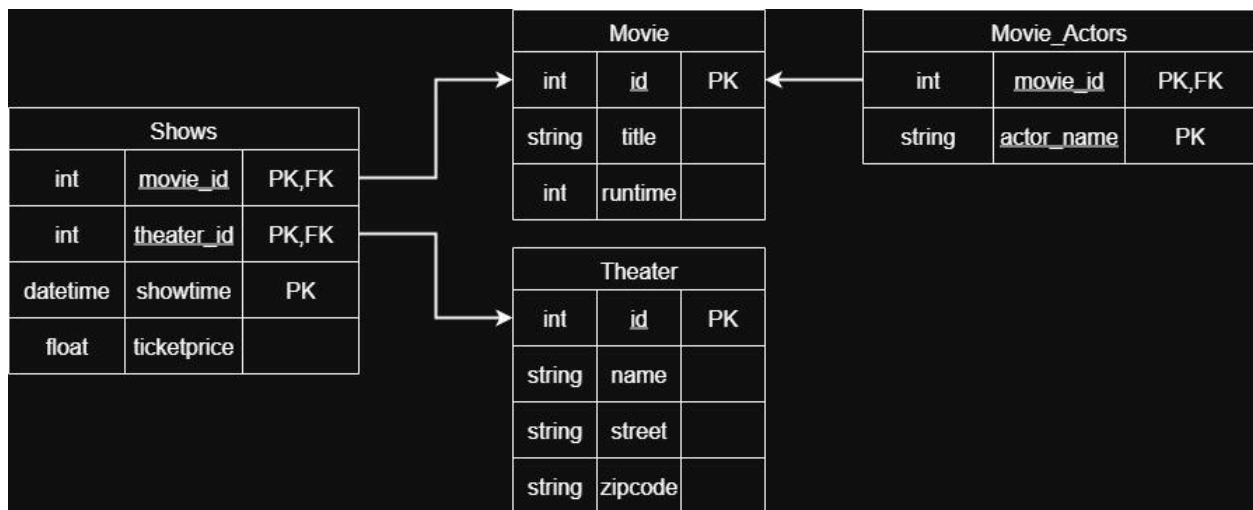


CS 2300 HW 3

Problem 1

Convert the following ER diagram into relational schemas. Make sure to denote all primary keys and foreign keys by drawing the arrows and include the data types for each attribute, e.g. R(id:int, name:str, ...).

Artificial keys should only be used when absolutely necessary. If a natural key exists and an artificial key is used instead, points will be deducted



Problem 2

Convert the following EER diagram into a relational database schema.

Figure 4.12 shows an example of an EER diagram for a small-private-airport database; the database is used to keep track of airplanes, their owners, airport employees, and pilots.

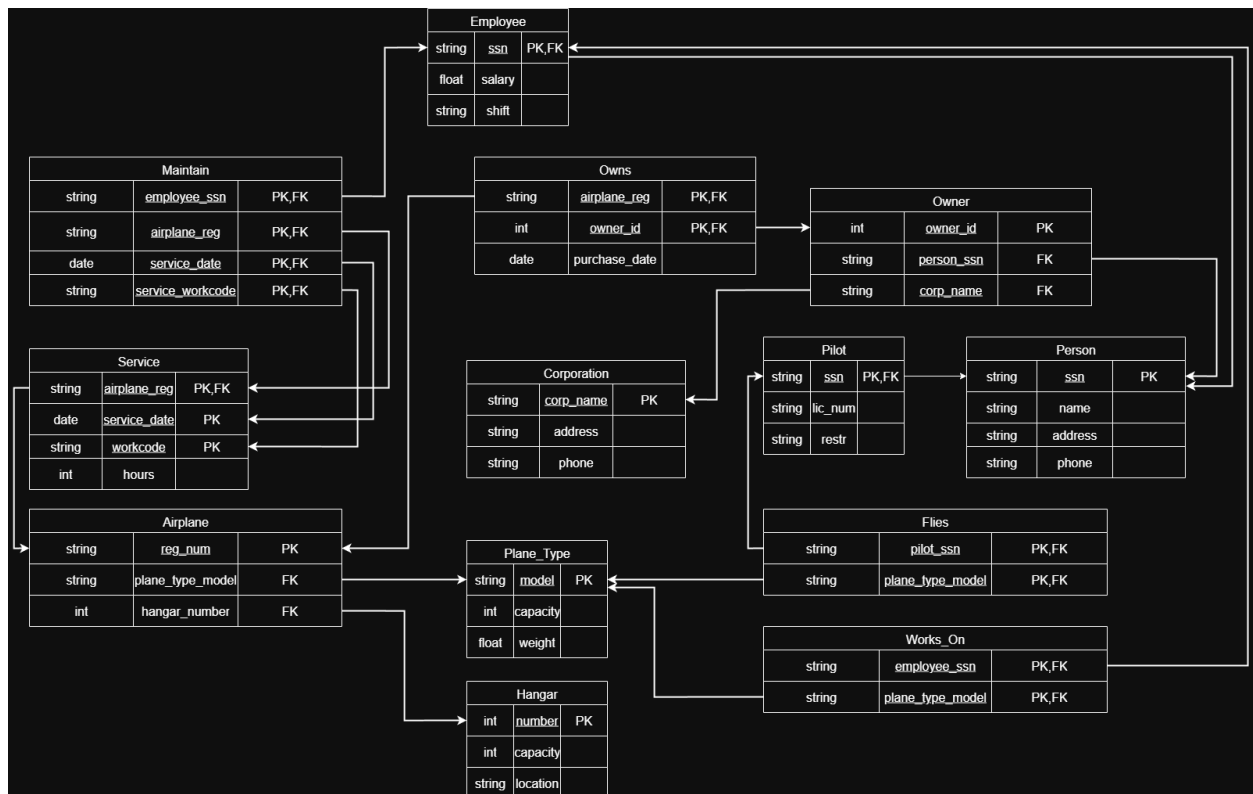
From the requirements for this database, the following information was collected:

- Each AIRPLANE has a registration number, is of a particular plane type, and is stored in a particular hangar.
- Each PLANE_TYPE has a model number, a capacity, and a weight.
- Each HANGAR has a number, a capacity, and a location.
- The database also keeps track of the OWNERS of each plane and the EMPLOYEES who have maintained the plane.
- Each relationship instance in OWNS relates an AIRPLANE to an OWNER and includes the purchase date.
- Each relationship instance in MAINTAIN relates an EMPLOYEE to a service record.
- Each plane undergoes service many times.
- A SERVICE record includes as attributes the date of maintenance, the number of hours spent on the work, and the type of work done.
- Airplane service is represented as a weak entity type because the airplane registration number is used to identify a service record.
- An OWNER is either a person or a corporation, thus represented using a union type.
- Both pilots and employees are subclasses of PERSON.
- Each PILOT has specific attributes license number and restrictions.
- Each EMPLOYEE has specific attributes salary and shift worked.
- All PERSON entities in the database have data kept on their Social Security number, name, address, and telephone number.
- For CORPORATION entities, the data kept includes name, address, and telephone number.

- The database also keeps track of the types of planes each pilot is authorized to fly and the types of planes each employee can do maintenance work on.

Make sure to denote all primary keys and foreign keys by drawing the arrows and include the data types for each attribute, e.g. R(id:int, name:str, ...).

Artificial keys should only be used when absolutely necessary. If a natural key exists and an artificial key is used instead, points will be deducted.



Problem 3

Convert the following EER diagram into a relational database schema.

The figure below shows an example of an EER diagram for the Japanese Vanisher Company. It is a secretive and highly specialized organization that helps individuals disappear from society for various reasons, whether they are escaping danger, starting a new life, or hiding from threats. The company provides clients with safe houses, new identities, and necessary survival tools while ensuring their tracks are erased. This database is designed to manage the logistics of these operations.

From the requirements for this database, the following information was collected:

- A Client is assigned a Vanisher to help them disappear.
- Both SafeHouse and DisappearanceKit belong to the Resource category/union type.
- A Client undergoes Operations, involving exactly ONE Client and at least ONE or MORE vanishers. A safe house is optional—some clients may disappear without using one.
 - Each operation must have a specific Date.
 - A client CANNOT have multiple operations on the same day.

